

# Modernizing Digital Banking Operations at a Retail Bank

Final Project | Systems Analysis & Design

ABC Bank | Digital Transformation Initiative

# Executive Summary

## THE PROBLEM

- Legacy systems slow onboarding to 5+ days
- Manual processes cause errors and delays
- Poor system integration across channels

## KEY INSIGHTS

- Cloud-native microservices approach
- API-first integration strategy
- Containerized deployment model

## RECOMMENDED ACTIONS

- 1 Migrate to IBM Cloud platform
- 2 Implement CI/CD pipelines
- 3 Deploy event-driven architecture
- 4 Modernize database layer
- 5 Establish real-time monitoring

# Introduction

## THE OPPORTUNITY

Digital transformation of ABC Bank's core operations to modernize customer-facing and back-office systems through a structured systems analysis approach.

## APPROACH

- UML and BPMN process modeling
- Data flow and architecture analysis
- Agile methodology with phased delivery

## KEY QUESTIONS

- ? How to reduce onboarding time by 60%?
- ? How to ensure 99.9% system uptime?
- ? How to meet regulatory compliance while modernizing?

# Objectives

**60%**

Reduce onboarding  
from 5 to 2 days

**99.9%**

System availability  
target uptime

**80%**

Transaction  
automation rate

**Real-time**

Fraud detection  
implementation

**PCI-DSS**

Full KYC and  
security compliance

**40%**

Customer satisfaction  
score improvement

# Project Charter

| Element            | Details                                       |
|--------------------|-----------------------------------------------|
| Project Name       | Digital Banking Modernization                 |
| Sponsor            | CTO, ABC Bank                                 |
| Scope              | Core banking, onboarding, loans, payments     |
| Objectives         | Modernize legacy systems, improve efficiency  |
| Major Deliverables | Cloud platform, microservices, CI/CD pipeline |
| Timeline           | 12 months, 3 phases                           |

# Stakeholder Register

| Stakeholder | Role      | Attitude   | Interest | Impact | Influence | Comm. Pref.  | Hours |
|-------------|-----------|------------|----------|--------|-----------|--------------|-------|
| CTO         | Sponsor   | Supportive | High     | High   | High      | Weekly brief | 9-5   |
| IT Director | Tech Lead | Neutral    | High     | High   | High      | Bi-weekly    | 9-6   |
| Branch Mgr  | End User  | Resistant  | Medium   | Medium | Medium    | Monthly      | 8-5   |
| Compliance  | Advisor   | Supportive | High     | High   | Medium    | Weekly       | 9-5   |
| CS Lead     | End User  | Supportive | Medium   | Medium | Low       | Training     | Shift |

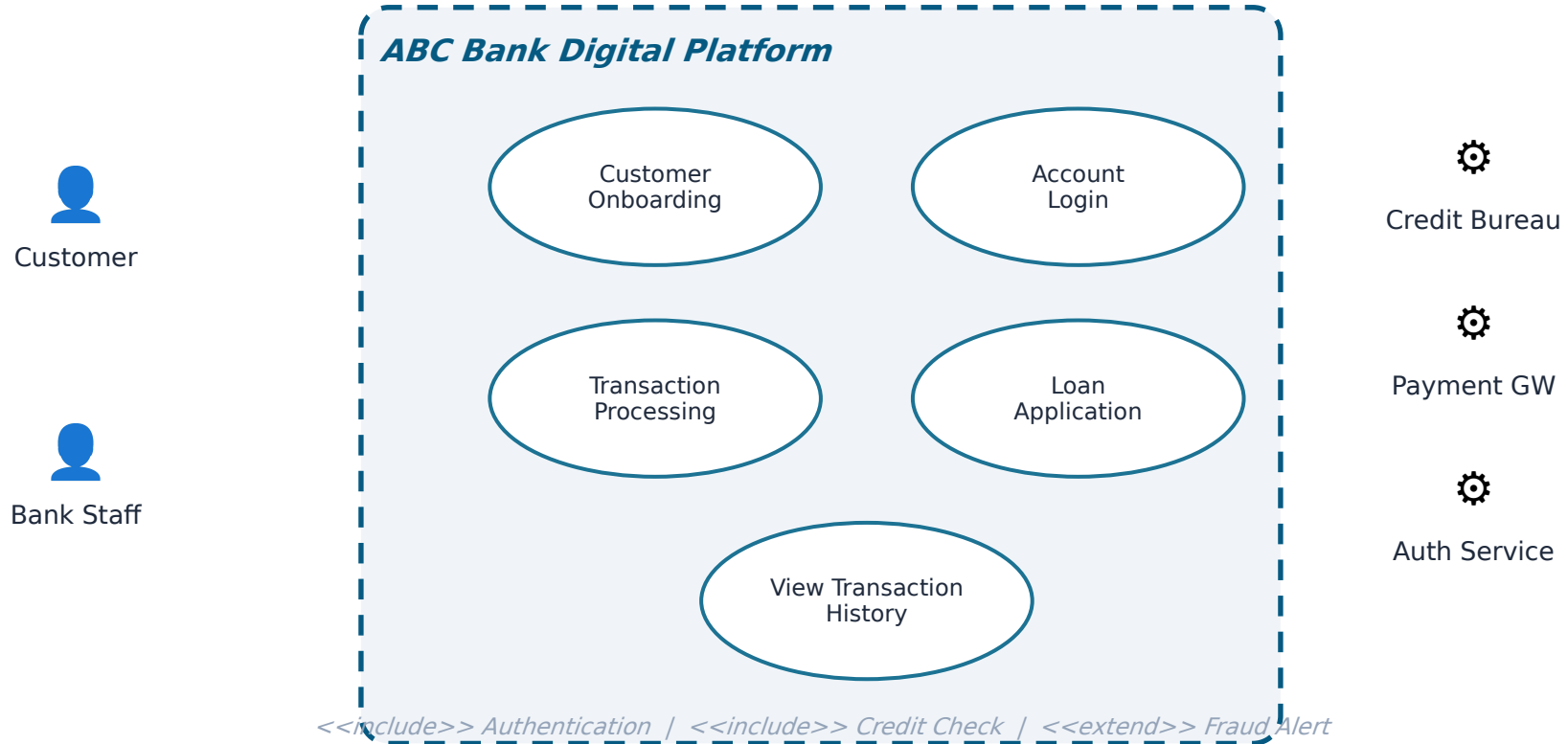
# Stakeholder Engagement Plan

| Stakeholder        | Current State | Desired State | Strategy                                  |
|--------------------|---------------|---------------|-------------------------------------------|
| CTO                | Leading       | Leading       | Regular progress demos and KPI dashboards |
| IT Director        | Neutral       | Supportive    | Involve in architecture decisions early   |
| Branch Manager     | Resistant     | Neutral       | Show workflow efficiency benefits early   |
| Compliance Officer | Supportive    | Leading       | Co-design compliance checks and audits    |
| CS Lead            | Supportive    | Supportive    | Provide training and feedback loops       |

# Business Requirements Document

| Section         | Details                                                                          |
|-----------------|----------------------------------------------------------------------------------|
| Scope           | End-to-end digital banking platform covering onboarding, transactions, and loans |
| Objectives      | Modernize onboarding, payments, and loan processing workflows                    |
| Assumptions     | Existing infrastructure supports hybrid cloud; staff available for training      |
| Success Metrics | Onboarding < 2 days   99.9% uptime   80% automation   Zero critical breaches     |

# Use Case Diagram



# Requirements Traceability Matrix

| Req ID  | Type | Description            | Source     | Acceptance Criteria | Owner     | Priority | Status      | Verification | Risk  |
|---------|------|------------------------|------------|---------------------|-----------|----------|-------------|--------------|-------|
| REQ-001 | Func | Online account opening | CTO        | Complete < 10 min   | IT Dir.   | High     | In Progress | UAT          | R-003 |
| REQ-002 | Func | Real-time transactions | Branch Mgr | Process < 2 sec     | Dev Lead  | High     | In Progress | Perf Test    | R-001 |
| REQ-003 | NFR  | 99.9% system uptime    | CTO        | < 8.7 hrs down/yr   | IT Dir.   | Critical | Planned     | Monitoring   | R-002 |
| REQ-004 | Func | Automated KYC          | Compliance | Verify < 5 min      | Comp Lead | High     | In Progress | Int. Test    | R-004 |
| REQ-005 | NFR  | PCI-DSS compliance     | Compliance | Pass annual audit   | Sec Lead  | Critical | In Progress | Sec Audit    | R-005 |

# Work Breakdown Structure

## 1.0 Digital Banking Modernization

### 1.1 Planning

1.1.1  
Requirements  
gathering

1.1.2  
Stakeholder  
analysis

1.1.3 Risk  
assessment

1.1.4  
Architecture  
design

### 1.2 Execution

1.2.1  
Infrastructure  
setup

1.2.2  
Microservices  
development

1.2.3 Data  
migration

1.2.4  
Integration  
testing

### 1.3 Closure

1.3.1 UAT

1.3.2  
Deployment

1.3.3  
Training

1.3.4 Post-  
launch  
review

# Network Diagram



*Critical Path: Requirements → Architecture → Infrastructure → Microservices → Integration Testing → UAT → Deployment*

Total Duration: ~32 weeks (8 months) | Planning: 2 wks | Execution: 6 months | Closure: 2 wks

# SWOT Analysis

## STRENGTHS

- Strong executive sponsorship
- Existing customer base
- Skilled IT team
- Clear regulatory framework

## WEAKNESSES

- Legacy system dependencies
- Limited cloud expertise
- Change resistance from staff
- Technical debt

## OPPORTUNITIES

- Cloud-native scalability
- AI-driven fraud detection
- Open banking APIs
- Market differentiation

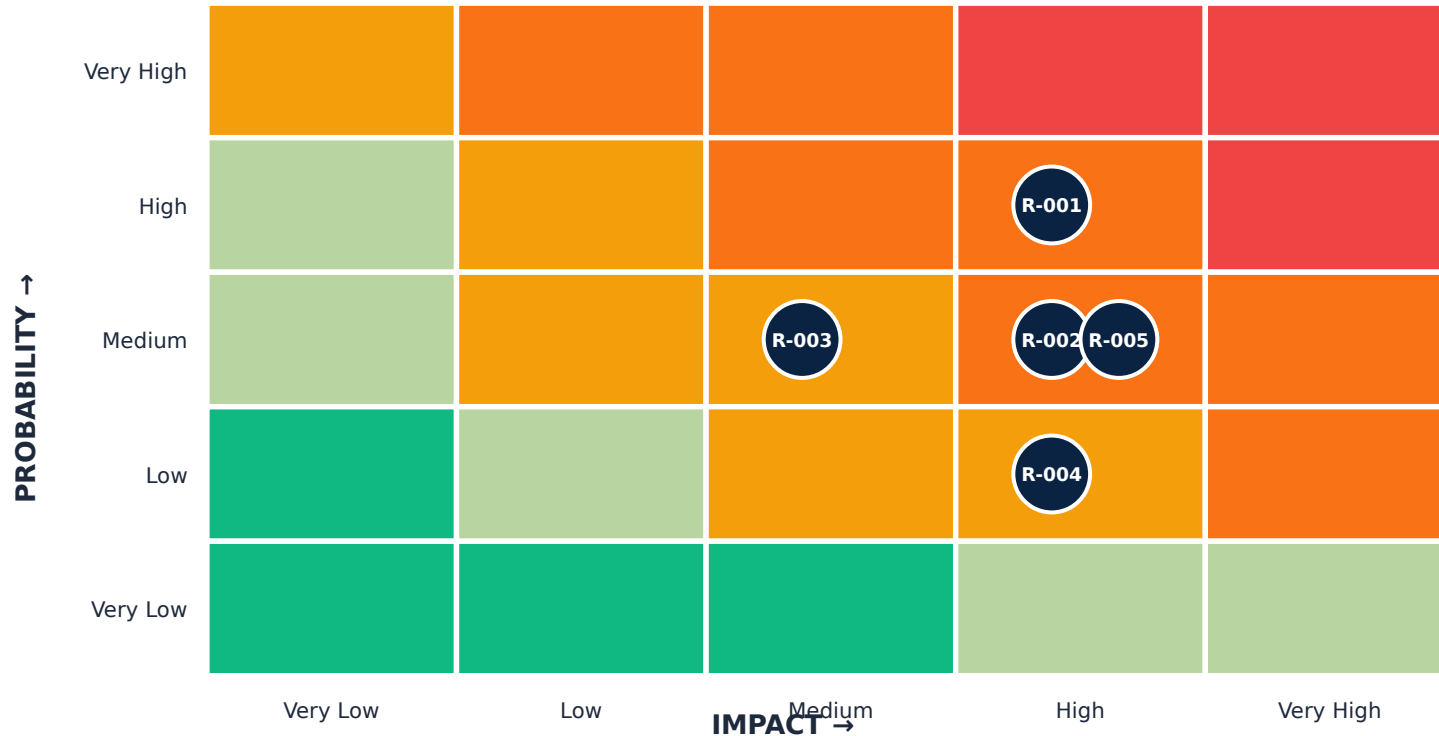
## THREATS

- Cybersecurity risks
- Regulatory changes
- Competitor digital offerings
- Data migration failures

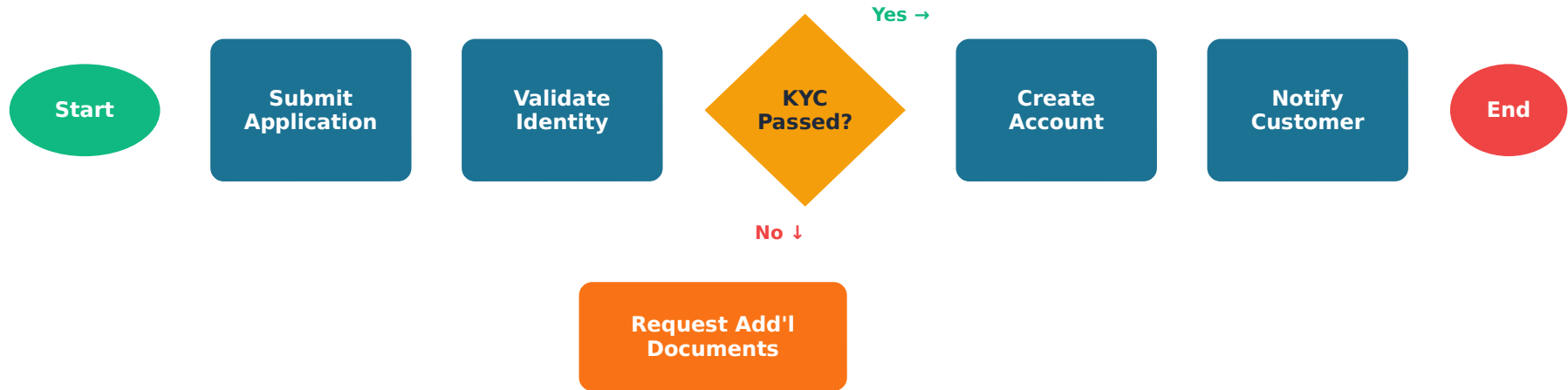
# Risk Register

| ID    | Cause             | Event           | Impact       | Owner    | Cat. | Prob | Impact | Rating   | Score | Response       |
|-------|-------------------|-----------------|--------------|----------|------|------|--------|----------|-------|----------------|
| R-001 | Legacy complexity | Migration fail  | Service down | IT Dir.  | Tech | High | High   | Critical | 9     | Rollback       |
| R-002 | Single point fail | Downtime        | Revenue loss | IT Dir.  | Tech | Med  | High   | High     | 6     | Failover       |
| R-003 | Manual process    | Onboard delay   | Cust. churn  | Br. Mgr  | Ops  | Med  | Med    | Med      | 4     | Automate       |
| R-004 | Reg. gaps         | Compliance fail | Fines        | Comp.    | Comp | Low  | High   | Med      | 3     | Audit          |
| R-005 | Cyber threats     | Data breach     | Reputation   | Sec Lead | Sec  | Med  | High   | High     | 6     | Incident resp. |

# Risk Matrix

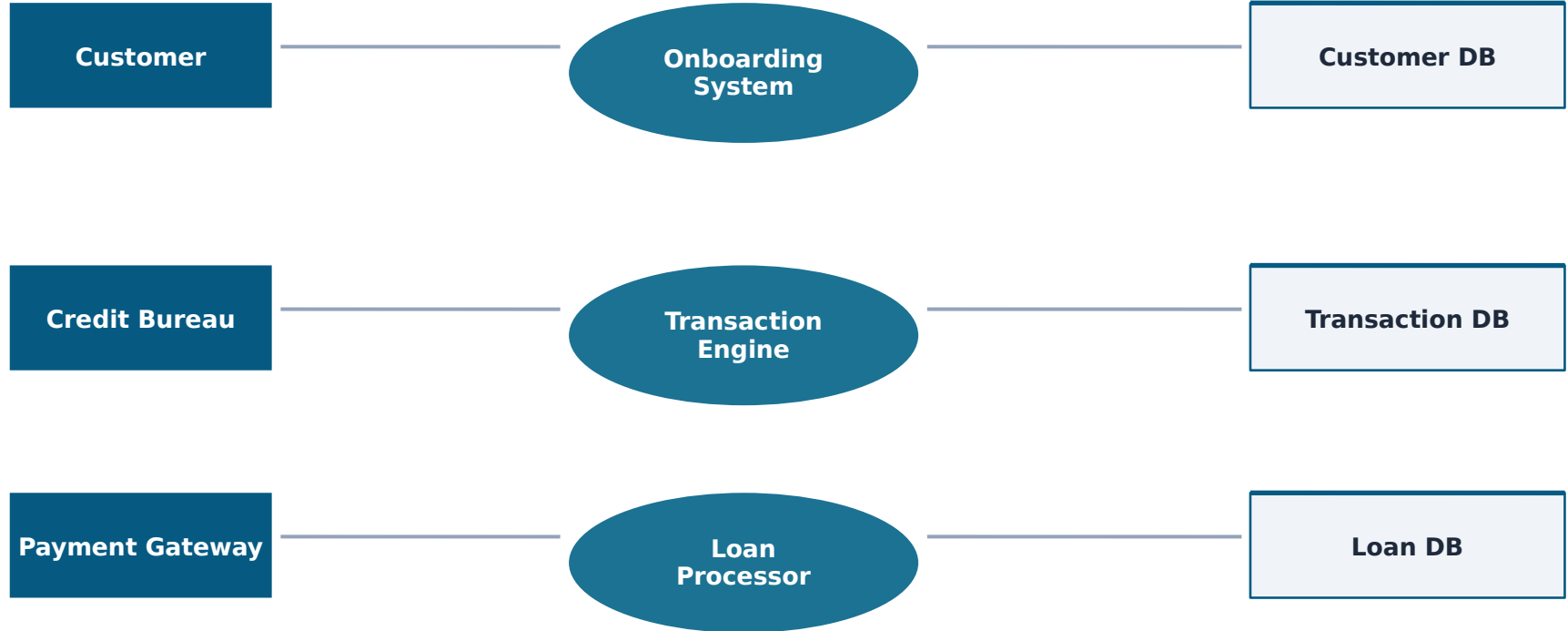


# Process Model: Customer Onboarding



*Key Components: Application Portal / Identity Verification / KYC Engine / Account Management / Notification Service*

# System Design: Data Flow Diagram (Level 0)



# System Architecture & Cloud Topology

## Presentation Layer

Web App

Mobile App

API Gateway

## Application Layer

Auth Service

Onboarding Svc

Transaction Svc

Loan Service

## Data Layer

PostgreSQL

Redis Cache

Kafka MQ

## IBM Cloud

Kubernetes Cluster

Container Registry

Load Balancer

WAF

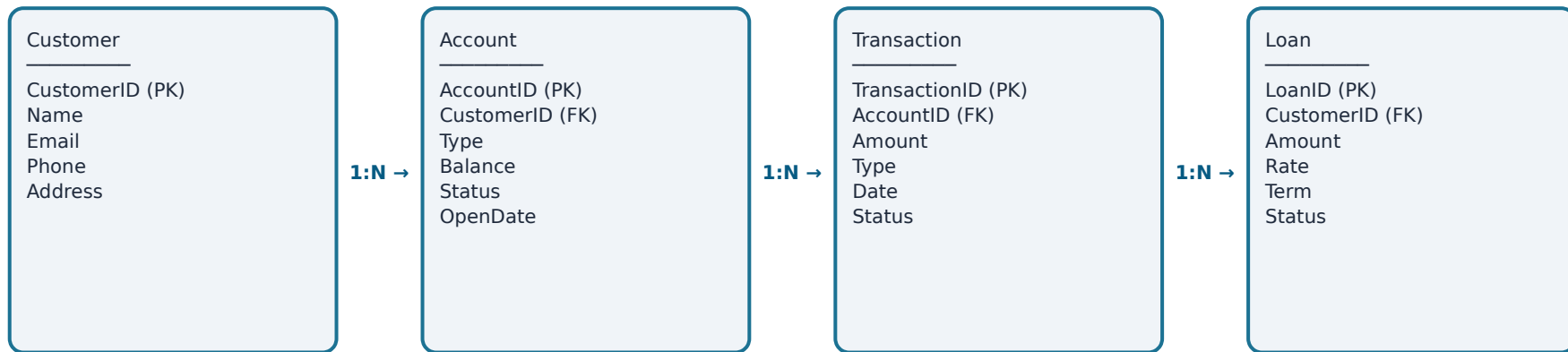
Cloud Monitoring

Cloud Logging

# Proof of Concept (PoC)

| Section         | Details                                                                |
|-----------------|------------------------------------------------------------------------|
| Objectives      | Validate microservices architecture for the customer onboarding flow   |
| Scope           | Auth service + Onboarding service + KYC integration                    |
| Setup           | Docker containers on IBM Cloud Kubernetes, PostgreSQL, REST APIs       |
| Tools           | Docker, Kubernetes, Jenkins, PostgreSQL, Kafka                         |
| Limitations     | Onboarding flow only; no production data; single region deployment     |
| Recommendations | Proceed with full build; add Redis caching; implement circuit breakers |

# Database Design



## Data Dictionary

| Field         | Type    | Length | Nullable | Description                 |
|---------------|---------|--------|----------|-----------------------------|
| CustomerID    | INT     | 11     | No       | Primary key, auto-increment |
| Name          | VARCHAR | 100    | No       | Customer full name          |
| AccountID     | INT     | 11     | No       | Primary key, auto-increment |
| Balance       | DECIMAL | 15,2   | No       | Current account balance     |
| TransactionID | INT     | 11     | No       | Primary key, auto-increment |

# Test Plan & Test Cases (TDD/BDD)

| Element     | Details                                                 |
|-------------|---------------------------------------------------------|
| Scope       | All microservices and integration points                |
| Purpose     | Validate functional and non-functional requirements     |
| Objectives  | 100% critical path coverage; performance benchmarks met |
| Approach    | TDD for unit tests; BDD for acceptance tests            |
| Environment | Staging cluster mirroring production                    |

## TDD Example

```
def test_account_creation():  
    account = create_account(valid_data)  
    assert account.status == "ACTIVE"  
    assert account.id is not None
```

## BDD Scenario (Gherkin)

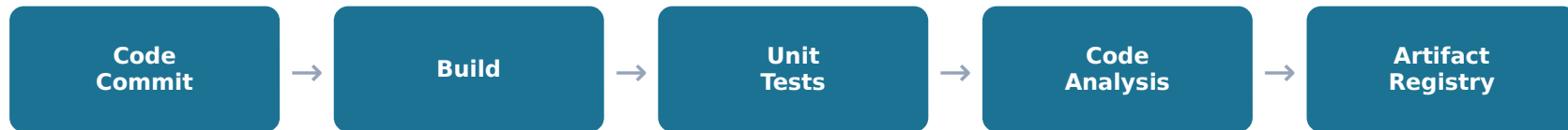
Feature: Customer Onboarding

Scenario: Successful account creation

Given a customer provides valid KYC documents  
When they submit the onboarding form  
Then an account is created within 10 minutes  
And a confirmation email is sent

# CI/CD Workflow & Monitoring Plan

## CI WORKFLOW



## CD PIPELINE



## MONITORING FRAMEWORK



# Key Findings

**300%**

## Onboarding Time Overhead

Legacy systems increase onboarding time by 300% compared to industry benchmarks due to manual data entry and paper-based verification.

**45 min**

## Incident Detection Delay

Lack of real-time monitoring means average incident detection takes 45 minutes, well above the 5-minute industry target.

**23%**

## Resource Waste

Current monolithic architecture prevents independent scaling, causing 23% resource waste during off-peak hours.

# Key Recommendations

1

## **Adopt Microservices Architecture**

Enable independent deployment and scaling of each banking service for greater agility.

2

## **Implement Event-Driven Architecture**

Use Kafka for real-time transaction processing and fraud detection across all channels.

3

## **Deploy CI/CD Pipeline**

Automated testing and deployment to reduce release cycles from monthly to weekly.

4

## **Establish Centralized Monitoring**

Prometheus and Grafana to achieve sub-5-minute incident detection and response.

# Conclusion

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- 1 The modernization initiative addresses critical operational inefficiencies in ABC Bank's digital infrastructure.
- 2 Cloud-native microservices architecture provides the scalability and resilience needed for modern banking.
- 3 Phased implementation minimizes risk while delivering incremental business value.
- 4 Success depends on stakeholder alignment, robust testing, and continuous monitoring post-deployment.

# Appendix

## REFERENCES

- IBM Cloud Documentation (2024)
- TOGAF Architecture Framework v10
- PCI-DSS Compliance Guide v4.0
- OWASP Security Testing Guide v4.2

## TOOLS USED

- Draw.io for UML and process diagrams
- Jira for project tracking and RTM
- Jenkins for CI/CD pipeline automation
- Prometheus/Grafana for monitoring

## SUPPORTING MATERIALS

- Detailed API specifications available in project repository
- Full stakeholder interview transcripts and meeting notes
- Complete test execution reports and defect logs